COMP90089: Machine Learning Applications for Health
Semester 2, 2026
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Group Project Assignment Instructions

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1 Introduction

So far in this subject, you have been introduced to the core concepts in health informatics, several different health data sources and their associated idiosyncrasies, the differences between routinely collected clinical data and research-ready datasets, digital phenotyping, diagnostic reasoning and their applications to modelling health-related problems. This material serves as background for the group project which makes up 50% of your total marks for this subject.

This is a group assignment, and groups have been allocated in Canvas. This group assignment forms the main summative assessment for the subject.

The assignment requires that you design, implement, and evaluate a machine learning project that addresses a relevant health problem.

The project has four assessed components:

Component	Mark type	Weight
Project proposal presentation	Group	10%
Project report	Group	20%
Project oral presentation	Group	10%
Peer evaluation	Individual	10%
Total		**50%**

The first three components receive a single mark shared by the whole group. The peer evaluation is an individual mark and is described in Section 5.

All submission deadlines are published on the Canvas assessment page, which is the authoritative source for dates. Note that the project oral presentations take place during the examination period, and attendance is required — please take this into account when making any travel plans.

2 Project Proposal

The first deliverable is an oral presentation of your project proposal, equivalent to approximately 500 words. This proposal will constitute 10% of your total marks for the subject. You will be required to:

- Briefly define the health problem your group chose and demonstrate its significance
- Describe the data processing and digital phenotyping approaches you intend to follow
- Describe the machine learning and evaluation strategies you will implement, including metrics of success

Regarding data, while your group is free to choose any health-related data source, given that MIMIC data is used extensively in the subject, it is expected that most groups will elect to utilise MIMIC as the primary data source for their assignment. However, if you decide to use a different dataset (i.e., not MIMIC) then that dataset should be publicly available. Groups using a non-MIMIC dataset will receive feedback on their choice at the proposal presentation.

For this project proposal, we ask you to answer the questions below, in a succinct way.

1. What is the clinical question/problem your group will tackle and why is it relevant? Please include information about the magnitude of the problem and make sure your claim is supported by references to the literature (up to 100 words).
2. Which data source(s) and information will you use? What is your plan for data processing and digital phenotyping? Remember, for a digital phenotype, it is important to list the criteria and describe which data elements you will use to find patients that meet those criteria (up to 200 words).
3. What is your machine learning approach and methodology? What are your metrics of success and expected outcomes? (up to 300 words)

Deliverable. The deliverable for the proposal is your slide deck, of up to 8 slides, including your citations. There is no separate written proposal document. The word counts above are the Handbook's rough notion of the size of the assignment, not a limit on what you say during the presentation.

The presentations will happen inside your tutorials and each group will have 8 minutes to present plus 2 minutes for questions and changeover to the next group. The order of the presentations will be randomly allocated. Note that the submission deadline for your slides is the same for all groups regardless of when your group is scheduled to present.

Our expectation is that you will cite at least five references in your proposal. Note that it does not matter which citation style you select, as long as it is consistent throughout. References may be randomly checked so please ensure the sources you cite exist and are relevant.

3 Project Report

The project report is worth 20% of your marks for this subject and should be approximately 2,000 words. Your report should:

1. Introduce the health problem your group has decided to tackle and its significance (building and expanding from your project proposal);
2. Describe the methodology your group has followed, including:
   (a) Data processing and digital phenotyping approaches you have followed
   (b) Machine learning and evaluation strategies you have implemented
3. Describe and discuss the obtained results, their implications, how they align with the literature and potential future steps.

Each group should include, at the end of their report, a table of contributions made by each group member using the CRediT — Contributor Role Taxonomy[1]

Note that in assessing your project, we are more interested in your critical analysis of methods and results than the raw performance of your models. It may be that you do not arrive at a definitive answer to your research question. However, you should analyse and discuss your (possibly negative) results in depth. You will be assessed on the quality of your report, not your code. However, your code should be sufficient to replicate the results provided in your report (i.e., in principle, it should be possible to run the code you supply to generate the results provided in your report). You are not required to implement algorithms from scratch. The use of existing library implementations of algorithms is encouraged.

Format

You should write the report using the JAMIA (Journal of the American Medical Informatics Association) format. A description of the format can be found here, but briefly, reports should consist of:

- **Structured abstract** — Write a short, structured abstract of up to 250 words containing the headings Objective, Materials and Methods, Results, Discussion, and Conclusion)
- **Introduction** — Provide a short description of the problem and data set, and the research question addressed. The Introduction section should include a summary of some related literature
- **Methods** — Explain the primary machine learning methods that you have used in your project with appropriate references. This section should include a description of your dataset and an ethics statement

[1] CRediT Taxonomy: https://credit.niso.org

- **Results** — Present your results in terms of evaluation metric(s) and, ideally, illustrative examples. Use of tables and figures is highly recommended
- **Discussion & Conclusion** — Contextualise and reflect on your results in the context of your research question. What have you learned? What are the limitations of your project?
- **References** — a specific referencing style is not required, but consistency in how you format references is important
- **Appendixes** — optional: any supplementary material that supports your claims but does not belong in the main text (for example additional tables, figures, or sensitivity analyses). Your code does not go here; it is submitted separately as a Jupyter notebook

The Methods section should include an ethics statement [use the heading **Ethics Statement**]. In most circumstances, ethics statements can be short and should typically discuss (a) whether the data is public and under which legal provisions have you accessed them; (b) any plans to redistribute data generated during the project; and (c) any sensitivity involved with the data.

Our expectation is that you will cite 15 to 30 references in writing your report. Note that it does not matter which citation style you select, as long as it is consistent throughout the report.

You can write the report using Microsoft Word, Pages, LaTeX, or Markdown. Templates are available for LaTeX[2] and Markdown[3]. Note that you can use as many tables and figures as you like. Please make sure that any figures used in the main body of the report are referenced in the text and discussed.

Word count

The report should be approximately 2,000 words, with the University's standard tolerance of ±10% (that is, 1,800–2,200 words). Content beyond 2,200 words will not be marked.

The following are **excluded** from the word count: the structured abstract (which has its own 250-word limit), the reference list, table and figure captions, the CRediT contributions table, the Ethics Statement, the generative AI declaration, and any appendixes. Everything else (Introduction, Methods, Results, Discussion and Conclusion) is included.

Assessment criteria

Your report will be assessed against the following criteria:

- Clarity of expression and cohesion

[2] LaTeX template: https://github.com/JinqianPan/JAMIA-paper-template
[3] Markdown template: https://github.com/erictleung/jamia-markdown-template

- Adequate contextualisation of the health problem in the literature
- Adequate use of visual aids (plots, figures, flowcharts, tables)
- Adequate methodological steps (i.e. sufficient detail to — in principle — replicate the work)
- Adequate discussion of results, implications, and future directions

Finally, in previous years we have seen many projects that run dozens of models and simply report which one performs best, with little analysis or discussion of the problem being studied. It is almost certainly a better strategy to evaluate fewer models — for example, a more interpretable model, a "black box" model, and a baseline model — combined with a deep analysis of the particular health-related problem you have selected.

4 Project Oral Presentation

During the examination period we will have oral presentations of all the projects. This presentation is worth 10% of your marks for the subject. As with the proposal, each group will have 8 minutes to present plus 2 minutes for questions and changeover.

The assessment rubric for both the proposal and the final oral presentation is published separately on Canvas.

You are expected to attend your fellow groups' presentations, out of respect for their work. Attendance is not itself marked.

5 Peer Evaluation

The peer evaluation is worth 10% of your marks for the subject. Unlike the other three components, this is an **individual** mark. It is completed on Canvas using FeedbackFruits.

The purpose of this component is to identify non-contribution, not to rank contributors. In a team where everyone has contributed fully, every member should expect full marks. It is not designed to differentiate between good and excellent teammates.

For each member of your team, including yourself, you will assign a score from 1 (very poor team member) to 5 (excellent team member) rating their contribution to the project, and justify that score with a brief statement describing what that person actually did. **A score submitted without a specific justification will be discarded rather than counted.**

Your own mark is derived from the average of the scores your teammates give you:

Average score received	Mark (out of 10)
4.0 or above	10
3.5 – 3.99	9
3.0 – 3.49	8
2.5 – 2.99	6
2.0 – 2.49	4
Below 2.0	0, following review

The following conditions apply:

- Individual scores and justifications are **confidential**. They are seen only by the teaching team and are never shown to your teammates.
- Submitted scores are cross-checked against the CRediT contributions table in your report.
- Where the average score received is below 2.0, the situation, not the person who gave the score, will be reviewed confidentially before any mark is assigned.
- Failure to submit your own peer evaluation means you do not get marks for this item.

Your group contract sets out how your team has agreed to work together and how to handle problems if they arise. Raise difficulties with your tutor early, while there is still time to resolve them. In cases of serious and documented non-contribution that were unable to be managed according to the group contract, the subject coordinator may adjust an individual's marks.

6 Uploading to Canvas

Please submit all files as PDFs (except the notebook) with your group number, the names of your group members, the student ID numbers of your group members, and your report's title clearly indicated on the title page. Please use page numbers in your report.

Replacing {GROUPNUMBER} with your group number, please use the following filenames:

Component	Deliverable	Filename
Proposal	Slides (max 8)	`ML4Health_proposal_slides_{GROUPNUMBER}.pdf`
Report	Report	`ML4Health_report_{GROUPNUMBER}.pdf`
	Slides	`ML4Health_report_slides_{GROUPNUMBER}.pdf`
	Notebook	`ML4Health_report_{GROUPNUMBER}.ipynb`
Peer evaluation	FeedbackFruits	Completed individually on Canvas

The Jupyter notebook must be in Python; Python is a prerequisite for this subject and other languages will not be accepted. The notebook is the code deliverable for this project. Do not also paste your code into the report.

7 Example Projects from Previous Year

Previous project titles include:

- Using blood glucose data to improve predictions of mortality for sepsis patients in the intensive care unit (ICU)
- Machine learning implementations on the COPD cohort of MIMIC-IV v2.2: K-means clustering and mortality prediction
- Diagnosing ventilator-associated pneumonia: a machine learning approach with MIMIC-IV data
- Predicting in-hospital mortality for ICU patients diagnosed with myocardial infarction: a machine learning approach using data from the first 24 hours of admission
- Mortality predictive analysis on acute pancreatitis in intensive care unit patients — MIMIC-IV dataset
- Prediction of adverse drug events due to drug-drug interactions in the intensive care unit
- Risk prediction modeling for ventilator-associated pneumonia (VAP) using early indicators in intensive care unit (ICU) patients

8 Academic Integrity

Every assignment will be checked for originality. The expectation is that every submission will be:

- Original work done by the author(s) of the assignment during this subject, during this semester;
- Appropriately referenced, meaning every claim needs a reference and references will be checked. You can pick the citation style but please be consistent.

We will apply the University's Academic Misconduct policy where plagiarism is deemed to have taken place.

Use of generative AI

Generative AI can be used for brainstorming, planning, proofreading and editing (as described here) but **must not be used for drafting text, generating code, or generating content**. Its use must be disclosed as per the University of Melbourne's current policy. **Every deliverable** (the report, the Jupyter notebook, and both slide decks) must include a declaration at the end stating:

- Whether a generative AI tool was used or not
- Which tool was used
- The purpose it was used for (brainstorming, planning, proofreading, or editing)
- An explicit declaration that it was **not** used for drafting/creating text or code, or for content generation

Penalties

- **Academic misconduct:** a penalty of 30% applies to the mark for the affected component, in addition to any action taken under the University's Academic Misconduct policy.
- **Late submission:** a penalty of 5% per day applies to the mark for the affected component, up to a maximum of 50%. If you have a reasonable explanation for a delay, you should apply for Special Consideration through the University's process rather than relying on this clause.
- **Moderation:** marks may be moderated so that the distribution of grades conforms to Faculty regulations.

9 Intended Learning Outcomes and Generic Skills

This project is associated with the following Intended Learning Outcomes:

- Describe the application of Artificial Intelligence concepts in the context of health problems
- Demonstrate familiarity with challenges associated with health data and data modelling
- Design, implement, and evaluate an AI system addressing a healthcare problem
- Critically assess Artificial Intelligence applications for health